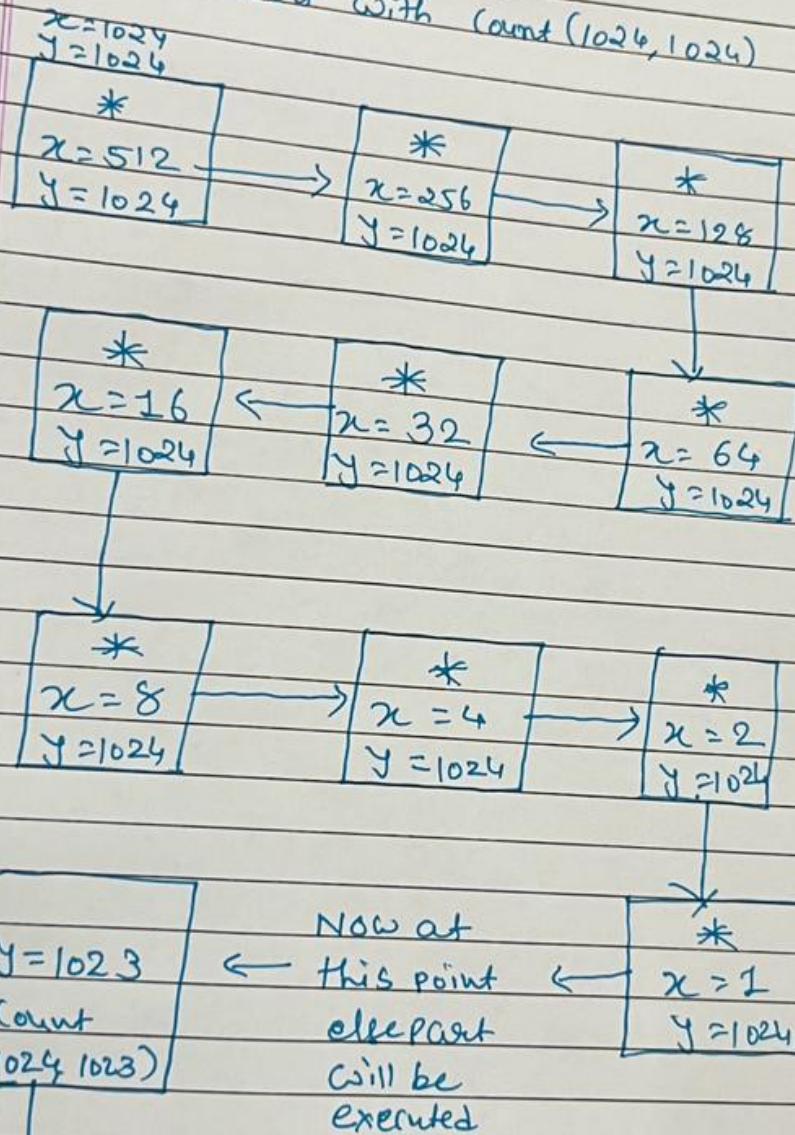


```
Count (x, y) {  
  if (y !=1 ) {  
    if (x !=1) {  
      print("*");  
      Count (x/2, y);  
    }  
    else {  
      y=y-1;  
      Count (1024, y);  
    }  
  }  
}
```

How many times * will be printed if we start with count (1024,1024)?

Let us start with count(1024, 1024)



This will
Recall & print *
until $x=1$.

So, starting from $x=1024$
to $x=1 \Rightarrow 10$ * gets
printed

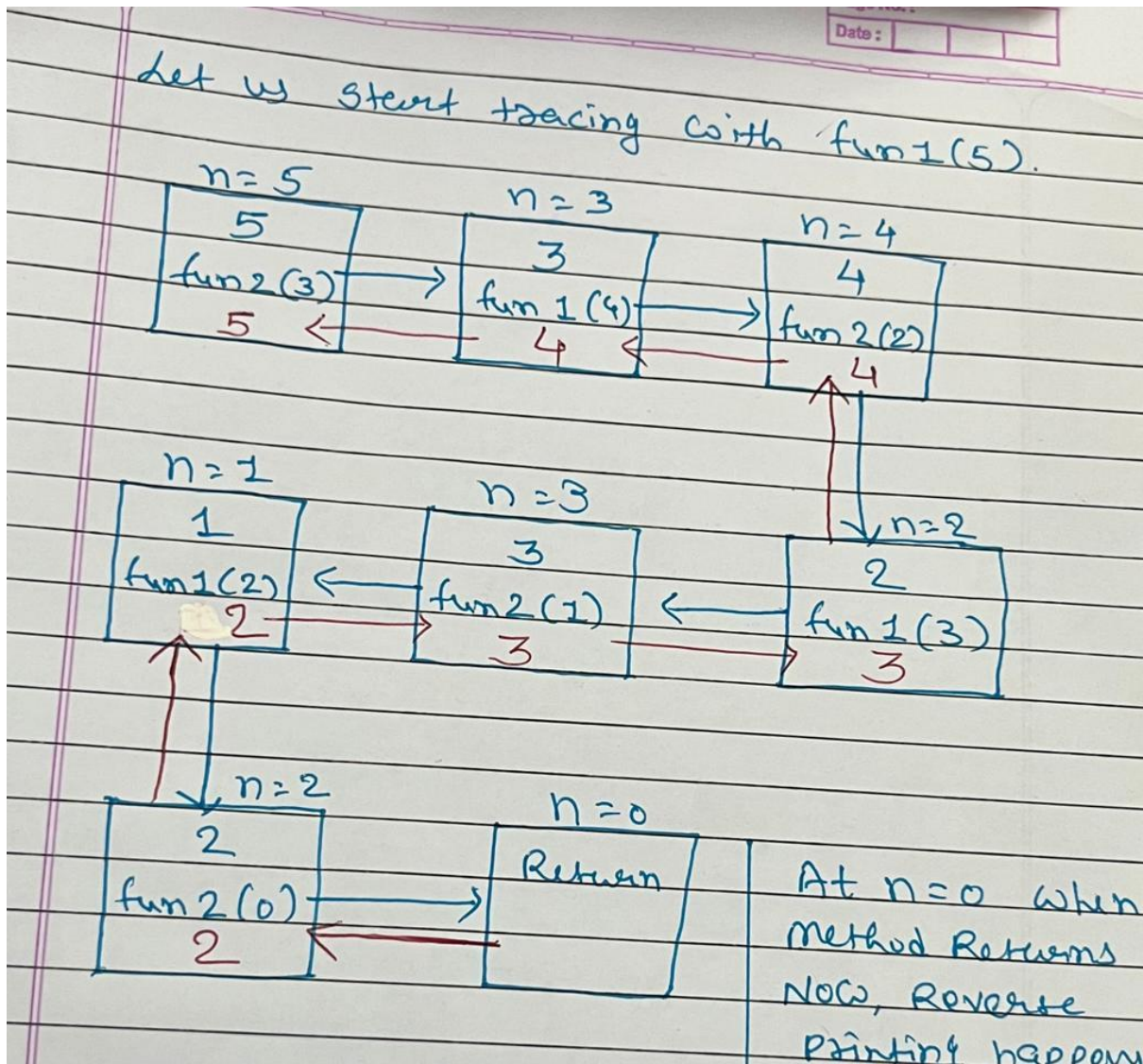
& at every $x=1$
 $y=y-1 \Rightarrow$ Initially $y=1024$
So, that goes up to 1023 times
 $\therefore * \Rightarrow 1023 \times 10 = 10230$
times.

```

1 void fun1(int n)
2 {
3     if (n==0)
4         System.out.print(n);
5         fun2(n-2);
6         System.out.print(n);
7 }
8 void fun2(int n)
9 {
10     if (n==0)
11         System.out.print(n);
12         fun1(++n);
13         System.out.print(n);
14 }

```

The output printed when fun1(5) is called is



So the blue color digits will be printed first as they are before the recursion calling.

So initial output will be 5 3 4 2 3 1 2 ---- Now at $n=0$ method returns to last recursion memory call and again prints 2 then goes back to the previous call but here due to $++n$ the n was updated to 3 instead of 1 so 3 will be printed next. Then again to previous call where $n = 3$ so again a 3 will be printed. So similarly while reverse tracking

The final output will be = 5 3 4 2 3 1 2 2 2 3 3 4 4 5